Domaine Scientifique de la DOUA
20, Avenue Albert Einstein - 69100 Villeurbanne

Industrial ecology and circular economy

Industrial ecology and circular economy

IDENTIFICATION

CODE : GI-4-S1-EC-EIE
ECTS : 2

HOURS

Cours :	0h
TD :	26h
TP :	0h
Projet :	0h
Evaluation :	2h
Face à face pédagogique :	28h
Travail personnel :	12h
Total :	40h

ASSESMENT METHOD

Written examination
Report

TEACHING AIDS

Powerpoint slides
Sustainable development and societal responsibility serious game (CIPE)

TEACHING LANGUAGE

French
English

CONTACT

M. ARBAOUI Taha :
taha.arbaoui@insa-lyon.fr

AIMS

This course belongs to teaching unit Continuous improvement and innovation(GI-4-S1-UE-AMCI) and contributes to the following skills :
A1 Analyze a system (real or virtual) or a problem (Level 2)
A2 Operate a model of a real or virtual system (Level 2)
C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
C3 Evaluating, prototyping and simulating a system (Level 2)
C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 1)
C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 2)
C12 Changing organizations to meet new constraints or opportunities (Level 2)
C20 Implementing Corporate Social Responsibility (CSR) (Level 1)
B5 act responsibly in a complex world (Level 2)
B6 position oneself, work, and evolve within a company or a socio-productive organization (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :
- Definitions of industrial ecology, circular economy and associated terminology (B5, B6)
- Material Flow Analysis (A1, A2, C1, C2, C3)
- Integrated Assessment Model (A1, A2, C1, C2, C3)
- Waste definition, associated waste management rules, legal aspects (A1, C10)
- Challenges of circular economy and reverse logistics from the industrial engineering point of view (C20, B5)
- Eco-parks, industrial symbiosis, synergy and mutualization (C11, C12)

To be able to :
- Propose solutions limiting the consumption and waste of resources and the production of waste (C10, C20)
- Diagnose sustainable development issues within an industrial company (C10, C11)

CONTENT

- Definitions
- Sustainable development and societal responsibility serious game
- Material Flow Analysis
- Best Available Techniques
- Waste
- Reverse logistics
- Ecoparks
- Disassembly problems

BIBLIOGRAPHY

PRE-REQUISITES

Notions of life cycle analysis